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Novel multi-functionalized fluorine-containing organometallics: Preparation and applications of tetrafluoroethylenated zinc reagent

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On treating 2,4-dibromo-3,3,4,4-tetrafluorobut-1-ene, readily prepared from commercially available 4-bromo-3,3,4,4-tetrafluorobut-1-ene (**1**), with 2.0 eq. of zinc-silver alloy at 40 °C in CH_3CN, zinc insertion at the terminal C(*sp*^{3})–Br site proceeded selectively, giving the corresponding (3-bromo-1,1,2,2-tetrafluorobut-3-en-1-yl)zinc bromide in high yield. This organozinc reagent was found to be thermally stable and easily handled. It was found that this organozinc reagent could participate in the cross-coupling reaction with various acyl chlorides or iodoarenes to give the corresponding CF_2CF_2-containing organic molecules in good yields. It was also revealed that Suzuki-Miyaura cross-coupling reactions at the C(*sp*^{2})–Br site of these CF_2CF_2-substituted products also proceeded in good to excellent yields.

1. Introduction

Since a fluorine atom has very specific properties which cannot be found in other elements, even the introduction of a small number of fluorine atoms into an organic substance can lead to dramatic changes in the molecular properties [1]. By taking advantage of such unique characteristics, fluorinated materials have gained a large share of the market in various fields, such as pharmaceuticals, agrochemicals and functional materials [2]. As a matter of course, a large number of synthetic chemists have considerably focused on the development of novel synthetic methods for the preparation of fluorinated compounds, and as a result, it becomes now possible to synthesize a wide variety of fluorinated organic compounds [3].

The most convenient and effective synthetic method for preparing fluorine-containing organic molecules is the one utilizing fluorine-containing organometallic reagents as a fluorinating agent after all [4]. In the case of multi-functionalized organometallic reagents, in particular, their versatility dramatically increases. This is because it is possible not only to introduce a fluorine-containing substituent into organic materials very easily but also to construct more complicated molecules through various chemical transformations, relying on the functional group(s) having in the organometallic reagents [5]. Despite such significant advantages, however, multi-functionalized fluorine-containing organometallic reagents have surprisingly been paid less attention [6], and hence their development is currently far behind.

In recent years, our research group has been energetically working on the study using organometallic reagents derived from commercially available 4-bromo-3,3,4,4-tetrafluorobut-1-ene (**1**) [7,8]. The reason why we have focused on the above study is because tetrafluoroethylene (CF_2CF_2)-containing materials have very promising applications in the fields of pharmaceuticals, agrochemicals and materials, since recent investigations have revealed that (1) tetrafluoroethylenated sugar derivatives have unique biological activities [9], (2) organic molecules having a CF_2CF_2-containing carbocycle in the mesogen part exhibit characteristics as negative liquid crystals [10]. In addition, another reason for focusing on the above research is that the organometallic reagents derived from **1** are very useful since they are "*functionalized fluorine-containing organometallic reagents*" with a carbon-carbon double bond moiety in addition to a carbon-metal bond.

We have succeeded in the transformation of **1** into various organometallic reagents, such as lithium (**2a**) [11], silicon (**2b**) [12], copper (**2c**) [13], and zinc (**2d**) reagents [14], so far, each of which has been found to possess very unique reactivities and be converted into a variety of CF_2CF_2-containing organic compounds (Scheme 1, Previous Works).

Therefore, we focused on novel tetrafluoroethylenated organozinc reagent, (3-bromo-1,1,2,2-tetrafluorobut-3-en-1-yl)zinc bromide (**3-Zn**), derived from 2,4-dibromo-3,3,4,4-tetrafluorobut-1-ene (**3**) which can be easily prepared from **1** [8a, 8b] (Scheme 1, This Work). This zinc reagent (**3-Zn**) is an unprecedented "*multi-functionalized fluorine-containing organometallic reagent*" with a carbon(*sp*^{2})–bromine

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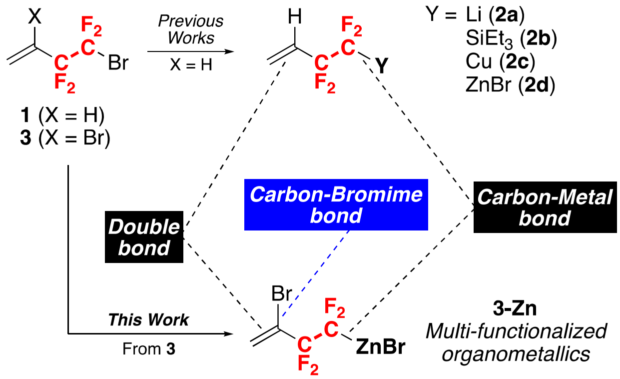


Scheme 1. Functionalized CF_2CF_2-containing organometallics.

bond, in addition to a carbon-metal bond and a carbon-carbon double bond contained in conventional CF_2CF_2-containing organometallic reagents (**2a**–**d**). Another major advantage is that all three functional groups can be chemically converted independently. Furthermore, this zinc reagent (**3-Zn**) can be expected to be thermally stable, just like **2-Zn**.

In this article, we describe the preparation of this novel zinc reagent (**3-Zn**) and the synthetic applications such as transition metal-catalyzed coupling reactions with various electrophiles and Suzuki-Miyaura cross-coupling reactions utilizing the carbon(sp^2)–bromine bond in detail.

2. Results and discussion

Initially, we started to investigate the preparation of zinc reagent having a tetrafluoroethylene group using dibromo compound (**3**) obtained in two steps from commercially available 4-bromo-3,3,4,4-tetrafluorobut-1-ene (**1**) [8a, 8b] (Table 1). First of all, we attempted to prepare **3-Zn** by applying the same reaction conditions used to prepare the zinc reagent **2d** Thus, the **3** was exposed to 2.0 equiv. of zinc-silver alloy Zn(Ag) in *N,N*'-dimethylformamide (DMF) at 0 °C, and the whole was stirred at room temperature for 1 h. As a result, the substrate **3** was consumed completely, and the corresponding zinc reagent **3-Zn** was obtained in only 48% yield (Entry 1) [14]. At this time, **3a** in which the terminal bromine was replaced with a hydrogen atom was obtained in 15% yield and **3b** in which the bromine of C(sp^2)–Br was also replaced with a hydrogen atom was obtained in 37% yield as a by-product, respectively. In addition, **4-Zn** by direct zinc insertion only into the C(sp^2)–Br bond was not obtained at all. Because of the above result, we reconsidered the preparation of novel organozinc reagent using **3** all over again from the beginning.

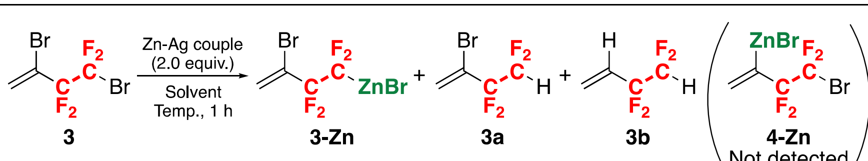
First, the reaction was attempted in various solvents (Entry 2–7). As a result, when diethyl ether was used as the solvent, the zinc insertion reaction did not proceed at all (Entry 2), but when acetonitrile (CH_3CN) was used, only **3-Zn** was obtained in a low yield while completely suppressing the formation of **3a** and **3b** (Entry 3). When 1,4-dioxane or dimethylsulfoxide (DMSO) was used, **3-Zn** was obtained only in a low yield. In addition, a large amount of starting material **3** was recovered, or a large amount of **3a** and **3b** were given as byproducts (Entry 4, 5). When *N,N*-dimethylacetamide (DMA) was used, **3a** and **3b** were also by-produced (Entry 6). Surprisingly, when THF was used, the reaction was dramatically improved and the desired **3-Zn** was obtained in 68% yield (Entry 7). At this time, the production of **3a** was completely suppressed, and **3b** was also produced only in 7% yield. Next, the reaction was attempted at 40 °C in diethyl ether, acetonitrile, 1,4-dioxane and THF. As a result, in the case of diethyl ether, 1,4-dioxane, and THF, the yield was not improved greatly (Entry 8, 10, 11), but when acetonitrile was used, the desired **3-Zn** was obtained in 88% yield, along with **3b** in 10% (Entry 9).

Now that we have established a highly efficient method for preparing zinc reagent **3-Zn**, we next attempted to the cross-coupling reaction of the **3-Zn** with various acid chlorides (Scheme 2). First, we applied the reaction conditions of the coupling between the zinc reagent **2d** and various acid chlorides for the reaction using **3-Zn**. Thus, treatment of 1.0 equiv. of **3-Zn** (0.5 M) with 2.4 equiv. of benzoyl chloride (**5a**) in the presence of 0.3 equiv. of Cu_2O and 0.2 equiv. of 2,2′-bipyridyl (bpy) in CH_3CN at room temperature for 20 h gave the desired product **6a** in only 3% yield. When the reaction was carried out using THF, DMF, *N,N*'-dimethylpropyleneurea (DMPU), and *N*-methylpyrrolidin-2-one (NMP) as a solvent instead of CH_3CN, an improvement of yield of **6a** was detected in a polar solvent, such as DMF, DMPU, and NMP. In particular, when the reaction was attempted using NMP as a solvent, the reaction proceeded almost quantitatively and the desired product **6a** was obtained in 96% yield.

With this reaction conditions, the coupling reactions of **3-Zn** with various acid chlorides were investigated (Scheme 2).

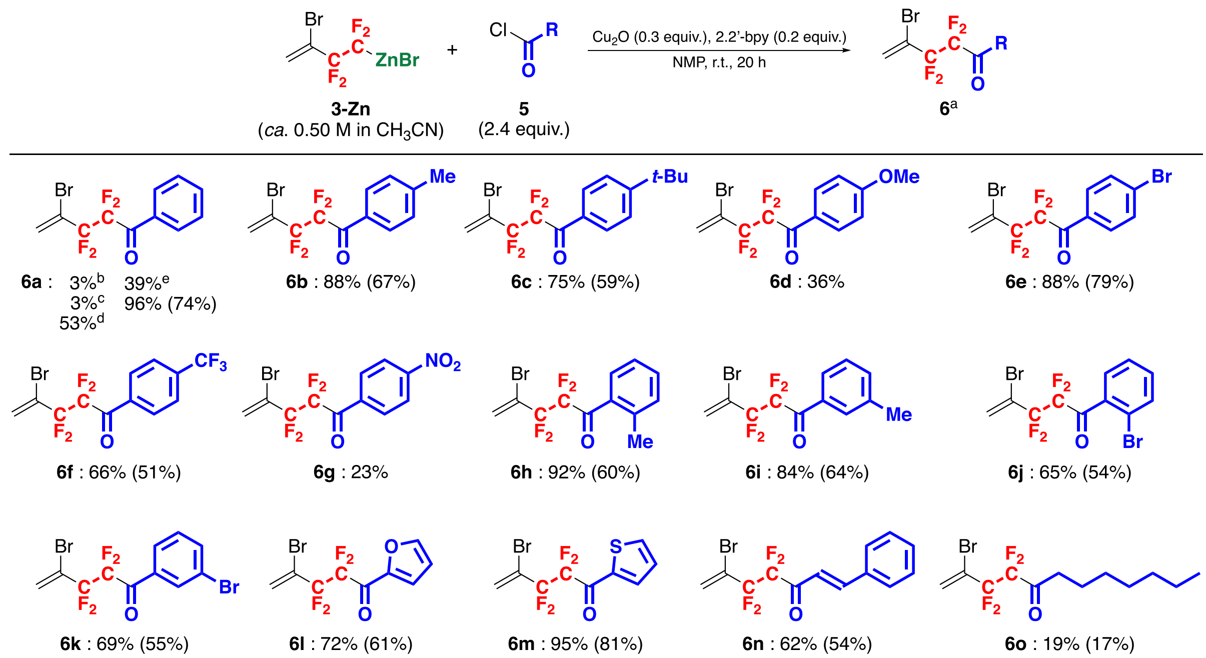
First, when benzoyl chloride derivatives having an electron-donating group, such as a methyl group or a *tert*-butyl group on the benzene ring, were employed, the reactions proceeded smoothly to give the corresponding coupling products **6b** and **6c** in good yield. Interestingly, it should be noted that the yield of **6d** decreased significantly when the benzoyl chloride having strongly electron-donating group, such as methoxy group on the benzene ring was employed. The reactions

Table 1
Screening of the reaction conditions for preparing the novel organozinc reagent.



Entry	Solvent	Temp./ °C	Yield [a]/% of **3-Zn**	Yield [a]/% of **3a**	Yield [a]/% of **3b**	Recovery [a]/% of **3**
1	DMF	r.t.	48	15	37	0
2	Et_2O	r.t.	0	0	0	100
3	CH_3CN	r.t.	5	0	0	95
4	1,4-Dioxane	r.t.	10	0	7	83
5	DMSO	r.t.	17	24	59	0
6	DMA	r.t.	50	11	39	0
7	THF	r.t.	68	0	7	0
8	Et_2O	40	19	0	0	63
9	CH_3CN	40	88	0	10	0
10	1,4-Dioxane	40	56	0	12	0
11	THF	40	57	0	6	0

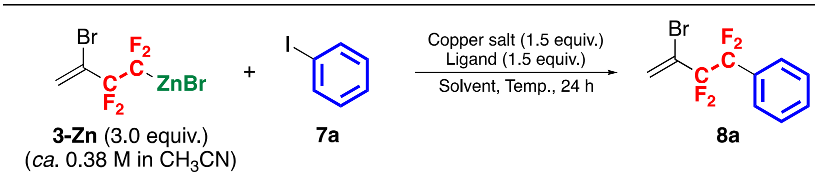
[a] Determined by ^{19}F NMR.



a) Yields are determined by ^{19}F NMR. Values in parentheses show isolated yields. b) Carried out in CH_3CN as solvent. c) Carried out in THF as solvent. d) Carried out in DMF as solvent. e) Carried out in DMPU as solvent.

Scheme 2. Cross-coupling with various acid chlorides.

Table 2
Optimization of the reaction conditions for the cross-coupling reaction of **3-Zn** with iodobenzene.



Entry	Solvent	Copper salt	Ligand	Temp./ °C	Yield[a]/%
1	CH_3CN	Cu_2O	2,2′-bpy	80	9
2	PhMe	Cu_2O	–	80	11
3	THF	Cu_2O	–	80	11
4	DMF	Cu_2O	–	80	6
5	DMA	Cu_2O	–	80	8
6	DMI	Cu_2O	–	80	43
7	NMP	Cu_2O	–	80	22
8	DMPU	Cu_2O	–	80	63
9	DMPU	CuI	–	80	50
10	DMPU	CuBr	–	80	56
11	DMPU	$CuBr_2$	–	80	51
12	DMPU	CuCl	–	80	50
13	DMPU	CuOAc	–	80	40
14	DMPU	CuSCN	–	80	50
15	DMPU	CuTC	–	80	66
16	DMPU	Cu_2O	1,10-phen	80	32
17	DMPU	Cu_2O	2,2′-bpy	80	33
18	DMPU	Cu_2O	TMEDA	80	53
19	DMPU	Cu_2O	Py	80	80
20	DMPU	Cu_2O	Py	60	18
21	DMPU	Cu_2O	Py	100	86
22[b]	DMPU	Cu_2O	Py	100	90 (28)[c]
23[d]	DMPU	Cu_2O	Py	100	31
24[e]	DMPU	Cu_2O	Py	100	56

[a] Determined by ^{19}F NMR. Value is parentheses shows isolated yield.
[b] Cu_2O (0.3 equiv.) and pyridine (0.3 equiv.) were used.
[c] The low isolated yield was due to high volatility of the product.
[d] Cu_2O (0.1 equiv.) and pyridine (0.1 equiv.) were used.
[e] Cu_2O (0.3 equiv.) and pyridine (0.3 equiv.) were used, and the whole was stirred for 16 h.

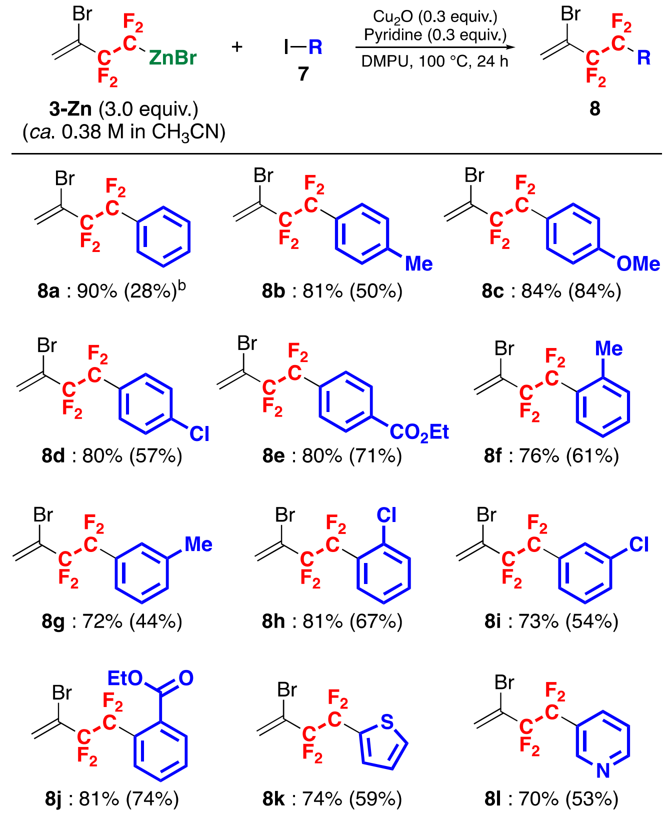
proceeded well to give corresponding products **6e** and **6f** in good yields when benzoyl chlorides having an electron-withdrawing group, such as a bromine substituent or a trifluoromethyl group on benzene ring was used instead of those having an electron-donating group. On the other hand, the yield of **6g** decreased significantly when the benzoyl chloride having a strongly electron-withdrawing group, such as a nitro group on the benzene ring was employed. As can be seen from the results when a methyl or a bromine substituent at the *ortho* or *meta* position of the benzene ring was used, it was found that the coupling product (**6h-k**) was obtained in good yields respectively regardless of the position of the substituent on the benzene ring in the benzoyl chloride derivatives. The reactions proceeded smoothly to give corresponding coupling products **6l** and **6m** in good yields when acid chlorides with a heteroaromatic moiety, such as 2-furoyl chloride (**5l**) and 2-thiophenecarbonyl chloride (**5m**) were used. In addition to aromatic acid chlorides, we also attempted aliphatic acid chlorides. As a result, in the case of cinnamoyl chloride (**5n**), the reaction proceeded without any problem, but when octanoyl chloride (**5o**) was used, the corresponding adduct (**6o**) was obtained in only 19% yield.

We next examined the cross-coupling of organozinc reagent **3-Zn** with various iodoarenes as further synthetic application.

First, the coupling reaction with iodobenzene was attempted with reference to the reaction conditions using acid chlorides described above (Table 2). Thus, treatment of 1.0 equiv. of iodobenzene with 3.0 equiv. of **3-Zn** (0.38 M) in the presence of 1.5 equiv. of Cu_2O and 1.5 equiv. of 2,2′-bipyridyl (bpy) in acetonitrile at 80 °C for 24 h afforded the desired product **8a** in only 9% yield (Entry 1). Therefore, the reaction using **3-Zn** and iodobenzene (**7a**) was re-examined to find out the optimal reaction conditions. First, when toluene, THF, DMF, DMA, 1,3-dimethyl-2-imidazolidinone (DMI), NMP, and DMPU were used as a solvent in the presence of 1.5 equiv. of Cu_2O (Entry 2–8), it was found that the yield was improved by using more polar solvent, such as DMI, NMP, and DMPU (Entry 6–8). In particular, when the reaction was attempted using DMPU as a solvent, the reaction proceeded smoothly and the desired product **8a** was obtained in 63% yield. Instead of Cu_2O, the reactions were conducted in various copper salts, such as CuI, CuBr, $CuBr_2$, CuCl, CuOAc, CuSCN, and CuTC, but no significant change was observed in the yield of the desired **8a**. In order to examine the ligand effect on the copper, next, the various nitrogen ligands were examined. When the bidentate ligands, such as 1,10-phenanthroline (1,10-phen), 2,2′-bipyridyl (2,2′-bpy) and TMEDA were used (Entry 16–18), the yields were not improved at all. Interestingly, the yield of **8a** was improved to 80% when pyridine was used as a monodentate ligand (Entry 19). Then, changing the reaction temperature from 60 °C to 100 °C caused a slight improvement of the yield, the **8a** was obtained in 86% yield (Entry 21). In the examination of reduction of copper salt and ligand, no change was observed in the yield when both were used in 0.3 equiv., but it was also revealed that the yield decreased significantly when equivalent number of both copper salt and ligand was reduced to 0.1 equiv. (Entry 23). We also found that the most suitable reaction time was 24 h (Entry 24).

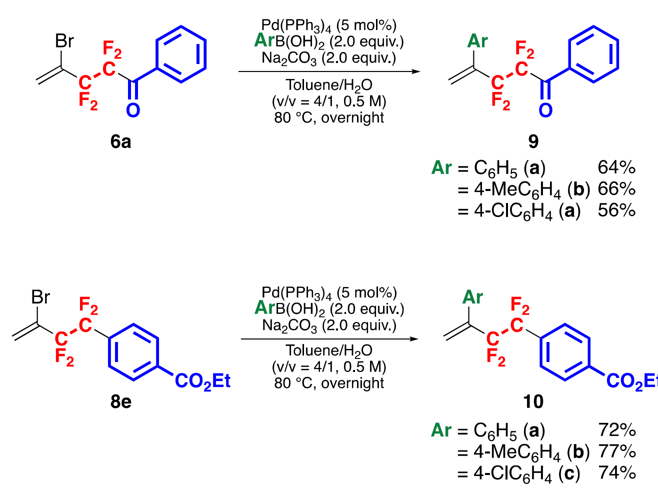
With the established optimum reaction conditions (Table 2, Entry 22), we next examined the coupling reactions of **3-Zn** with various iodoarenes **7** (Scheme 3). *para*-Iodotoluene (**7b**), *para*-iodoanisole (**7c**), *para*-chloroiodobenzene (**7d**), and ethyl *para*-iodobenzoate (**7e**), which has an electron-donating or an electron-withdrawing group on the benzene ring all participated in the cross-coupling reaction very well to give the corresponding coupling products (**8b**−**e**) in high yields. As can be seen in the cross-coupling reactions of *ortho*- or *meta*-iodotoluenes (**7f**, 7 g), *ortho*- or *meta*-chloroiodobenzenes (**7h, 7i**), and ethyl *ortho*-iodobenzoate (**7j**), the reactions proceeded smoothly regardless of the position of the substituent on the benzene ring. Heteroarenes, such as 2-iodothiophene (**7k**) or 3-iodopyridine (**7l**), were also fully applicable to this reaction, affording the corresponding coupling products (**8k, 8l**) in good yields.

Next, the Suzuki-Miyaura cross-coupling reaction [15] was attempted utilizing the C(sp^2)–Br bond part of thus-obtained coupling products



a) Determined by ^{19}F NMR. Values in parentheses show isolated yields.
b) The low isolated yield is due to high volatility of the product.

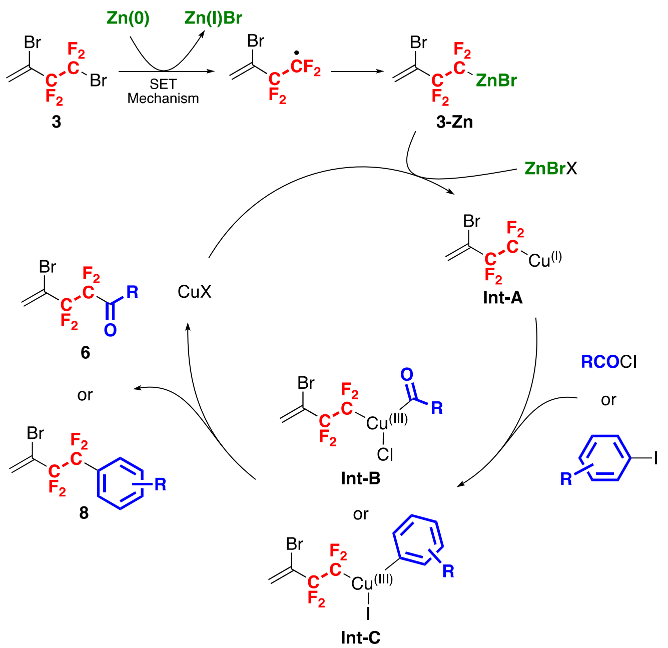
Scheme 3. Cross-coupling reaction of **3-Zn** with various iodoarenes.



Scheme 4. Suzuki-Miyaura cross-coupling reactions of **6a** or **8e**.

6a and **8e** as a further synthetic application (Scheme 4). Treatment of 1.0 equiv. of **6a** or **8e** with 2.0 equiv. of various arylboronic acids in the presence of 5 mol% of $Pd(PPh_3)_4$ and 2.0 equiv. of Na_2CO_3 in toluene/H_2O as a mixed solvent at 80 °C overnight gave the corresponding CF_2CF_2-containing 1,1-disubstituted alkenes **9a**–**c** and **10a**–**c** in good yields, respectively. The synthetic sequence using the multi-functionalized organometallics would become promising approaches to produce a variety of organofluorine compounds containing a CF_2CF_2-fragment.

A possible reaction mechanism for the cross-coupling reaction of **3-Zn** with various acid chlorides or iodoarenes is shown in Scheme 5 [14,



Scheme 5. Possible reaction mechanism for the cross-coupling of **3-Zn** with **6** and **8**.

16].

That is to say, at first, a fluoroalkyl radical is generated by a single electron transfer (SET) from zinc to **3**, and then recombines with monovalent zinc to generate a zinc reagent **3-Zn** [5 g, 17]. The preferential insertion of zinc into the terminal carbon(sp^3)–bromine bond is due to high stability of the intermediate perfluoroalkyl radical species. Next, transmetallation of **3-Zn** with a copper(I) salt gives tetrafluoroethylene-containing alkyl copper intermediate (**Int-A**), and then the Cu(I) atomic center oxidatively add to the $C - X$ (X = Cl or I) bonds of various acid chlorides or iodoarenes, giving rise to the copper (III) intermediate (**Int-B** or **Int-C**). Finally, reductive elimination of the **Int-B** or **Int-C** gives the corresponding **6** and **8** along with the regeneration of the Cu(I) salt.

3. Conclusions

In summary, treatment of 2,4-dibromo-3,3,4,4-tetrafluorobut-1-ene which can be readily prepared from commercially available 4-bromo-3,3,4,4-tetrafluorobut-1-ene (**1**), with zinc-silver alloy resulted in the selective insertion of zinc at the terminal C(sp^3)–Br site, giving a novel tetrafluoroethylenating agent, *(3-bromo-1,1,2,2-tetrafluoro-3-en-1-yl) zinc bromide (**3-Zn**)*. By the reaction of this reagent with 2.4 equiv. of various acid chlorides in the presence of 0.3 equiv. of copper oxide and 0.2 equiv. of 2,2′-bipyridyl in CH_3CN/NMP mixture and stirring at room temperature for 20 h, the corresponding ketones were obtained in high yields. In the presence of 0.3 equiv. of Cu_2O and 0.3 equiv. of 2,2-bipyridyl in CH_3CN/DMPU mixture, 1.0 equiv. of various iodoarenes were treated with 3.0 equiv. of the **3-Zn** (0.5 M) at 100 °C for 24 h, affording the corresponding aromatic compounds in high yields as well. Furthermore, the Suzuki-Miyaura cross-coupling reaction utilizing the C(sp^2)–Br site also proceeded extremely smoothly. From the above results, it could be proved that **3-Zn** was very useful as a multi-functionalized CF_2CF_2-containing organometallic reagent.

4. Experimental

4.1. General experimental

Infrared spectra (IR) were determined in a liquid film on a NaCl plate with a JASCO FT/IR-4100 type A spectrometer and all spectra are reported in wavenumber (cm^{-1}). ^{1}H and ^{13}C NMR spectra were measured with a Bruker AVANCE III 400 NMR spectrometer (^{1}H: 400 MHz and ^{13}C: 100 MHz) in a chloroform-*d* ($CDCl_3$) solution and the chemical shifts are reported in parts per million (ppm) using the residual proton or carbon in the NMR solvent. A Bruker AVANCE III 400 NMR spectrometer was used for determining the yield of the products with hexafluorobenzene (C_6F_6) or benzotrifluoride ($CF_3C_6H_5$). ^{19}F NMR (376.50 MHz) spectra were measured with a Bruker AVANCE III 400 NMR spectrometer in a $CDCl_3$ solution with trichlorofluoromethane ($CFCl_3$) as an internal standard. High-resolution mass spectra (HRMS) were taken on a JEOL JMS-700MS spectrometer by fast atom bombardment (FAB) method.

All reactions were carried out using dried glassware with a magnetic stirrer bar under an atmosphere of argon and routinely monitored by ^{19}F NMR spectroscopy or thin-layer chromatography (TLC).

All chemicals were of reagent grade and, if necessary, were purified in the usual manner prior to use. Column chromatography was carried out on silica gel (Wako gel® C-200) and TLC was performed on silica gel TLC plates (Merck, Silica gel 60 F_{254}).

4.2. Preparation of Zn-Ag alloy

To a stirred solution of silver acetate (10 mg, 0.060 mmol) in glacial acetic acid (20 mL) at reflux temperature was added zinc powder (10 g, 153 mmol) in one portion, and the whole was stirred at that temperature for additional 30 s. After refluxing for 30 s, the flask was cooled by dipping in ice-bath. The Zn-Ag couple formed was separated by vacuum filtration, which was washed with Et_2O several times, followed by drying under reduced pressure. The Zn-Ag alloy separated was stored at room temperature under Ar atmosphere.

4.3. Preparation of $CH_2{=}CBrCF_2CF_2Br$

To a solution of 4-bromo-3,3,4,4-tetrafluoro-1-butene (125.29 g, 0.625 mol) in dichloromethane (625 mL) was added dropwise bromine (32.2 mL). The solution was stirred at room temperature overnight. The reaction mixture was quenched with $NaHCO_3$ aq. and $Na_2S_2O_3$ aq. and exacted with dichloromethane three times. The organic layers were dried over anhydrous Na_2SO_4 and concentrated *in vacuo*. In the residue, the NaOH aq. (3 M) was added to 1,2,4-tribromo-3,3,4,4-tetrafluorobutane (210.90 g, 0.575 mol) and the mixture was stirred at 30 °C for 48 h. The whole was extracted. The organic layer was dried over anhydrous Na_2SO_4 and filtered to give the dibrominated compound (155.77 g, 90%).

4.3.1. 2,4-Dibromo-3,3,4,4-tetrafluorobut-1-ene (3)

Yield: 90%; Colorless oil; ^{1}H NMR ($CDCl_3$): δ 6.25 (dt, J = 3.0, 1.3 Hz, 1H), 6.53 (d, J = 2.9 Hz, 1H); ^{13}C NMR (100 MHz, $CDCl_3$): δ 111.6 (tt, J = 257.5, 31.5 Hz, CF_2), 116.7 (tt, J = 314.0, 42.3 Hz, CF_2), 116.9 (t, J = 29.4 Hz), 128.4 (t, J = 6.0 Hz); ^{19}F NMR ($CDCl_3$, $CFCl_3$): δ –63.41 (t, J = 4.9 Hz, 2F), –105.98 (t, J = 4.9 Hz, 2F); IR (neat): 1627, 1398, 1239, 1158, 1064, 1040, 934, 910, 870, 758, 681, 472 cm^{-1}; MS (FAB): *m/z* 155 (CF_2CBrCH_2, 38).

4.4. Preparation of $CH_2{=}CBrCF_2CF_2ZnBr$ in CH_3CN

In a two-necked round-bottomed flask, equipped with a teflon-coated stirrer bar, was placed Zn-Ag alloy (1.31 g, 10 mmol) in CH_3CN (20.0 mL). To the suspension was slowly dropwise 2,4-dibromo-3,3,4,4-tetrafluorobut-1-ene (2.858 g, 10 mmol) at 40 °C and the whole was stirred at that temperature for 1 h. The resultant was subjected to a short pad of celite and washed by Et_2O three times. The filtrate was concentrated *in vacuo* to give the corresponding organozinc reagents **3-Zn** in 88% NMR yield as a 0.50 M CH_3CN solution.

4.4.1. (3-Bromo-1,1,2,2-tetrafluorobut-3-en-1-yl)zinc bromide (3-Zn) in CH_3CN

Yield: 88% (determined by ^{19}F NMR); ^{19}F NMR ($CDCl_3$): δ –118.83 (s, 2F), –104.75 (s, 2F).

4.5. Typical procedure for Cu(I)-catalyzed cross-coupling reaction of organozinc reagent 3-Zn with benzoyl chloride

In a two-necked round-bottomed flask, equipped with a teflon-coated stirrer bar, were placed benzoyl chloride (0.15 mL, 1.3 mmol), Cu_2O (0.024 g, 0.17 mmol), 2,2′-bipyridyl (0.017 g, 0.11 mmol), NMP (1 mL) and a CH_3CN solution of **3-Zn** (0.55 M, 1.0 mL, 0.55 mmol), and the whole was stirred at room temperature for 20 h. The resultant was poured into water and the organic products were extracted with Et_2O three times. The combined organic layers were dried over anhydrous Na_2SO_4, then filtered, and concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give the corresponding **6a** (0.13 g, 0.41 mmol) in 74% yield as a pale yellow oil.

4.5.1. 4-Bromo-2,2,3,3-tetrafluoro-1-phenylpent-4-en-1-one *(6a)*

Yield: 74% (0.13 g, 0.41 mmol); Pale yellow oil, (Hexane/EtOAc = 50/1 R_f = 0.22); ^{1}H NMR ($CDCl_3$): δ 6.20 (dt, J = 2.8, 1.3 Hz, 1H), 6.51 (d, J = 2.8 Hz, 1H), 7.47–7.57 (m, 2H), 7.68 (tt, J = 7.5, 1.5 Hz, 1H), 8.10 (d, J = 7.5 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 111.8 (tt, J = 267.3, 37.1 Hz), 112.54 (tt, J = 256.4, 31.7 Hz), 117.8 (t, J = 28.8 Hz), 127.2 (t, J = 6.3 Hz), 128.7, 130.2 (t, J = 3.4 Hz), 132.2, 134.9, 185.0 (t, J = 26.2 Hz); ^{19}F NMR ($CDCl_3$, $CFCl_3$): δ –112.01 (s, 2F), –107.65 (s, 2F); IR (neat): 1704, 1625, 1598, 1579, 1450, 1286, 1158, 1078, 934, 758 cm^{-1}; HRMS (FAB) Calcd for $C_{11}H_8BrF_4O$ $[M+H]^+$: 310.9695, Found 310.9697.

4.5.2. 4-Bromo-2,2,3,3-tetrafluoro-1-(4-methylphenyl)pent-4-en-1-one *(6b)*

Yield: 67% (0.11 g, 0.33 mmol); Pale yellow oil (Hexane/Benzene = 10/1, R_f = 0.28); ^{1}H NMR ($CDCl_3$): δ 2.44 (s, 3H), 6.19 (dm, J = 2.9 Hz, 1H), 6.49 (d, J = 3.0 Hz, 1H), 7.31 (d, J = 8.2 Hz, 2H), 8.00 (d, J = 8.1 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 21.9, 112.0 (tt, J = 267.2, 36.9 Hz), 112.7 (tt, J = 256.6, 32.7 Hz), 118.3 (t, J = 29.0 Hz), 127.3 (t, J = 5.5 Hz), 129.7, 130.0, 130.7, 146.5, 184.8 (t, J = 26.0 Hz); ^{19}F NMR ($CDCl_3$): δ –111.97 (s, 2F), –107.67 (s, 2F); IR (neat): 2925, 1698, 1607, 1413, 1289, 1227, 1158, 1078, 884, 738 cm^{-1}; HRMS (FAB) Calcd for $C_{12}H_9BrF_4O$ $[M+H]^+$: 324.9851, found 324.9843.

4.5.3. 4-Bromo-1-(4-tert-butylphenyl)−2,2,3,3-tetrafluoropent-4-en 1-one *(6c)*

Yield: 59% (0.12 g, 0.31 mmol); Colorless oil (Hexane only, R_f = 0.25); ^{1}H NMR ($CDCl_3$): δ 1.36 (s, 9H), 6.20 (m, 1H), 6.50 (d, J = 2.9 Hz, 1H), 7.53 (dm, J = 8.7 Hz, 2H), 8.05 (d, J = 8.4 Hz 2H); ^{13}C NMR ($CDCl_3$): δ 30.9, 35.4, 112.0 (tt, J = 267.2, 37.0 Hz), 112.7 (tt, J = 256.3, 31.7 Hz), 118.2 (t, J = 28.8 Hz), 126.0, 127.3 (t, J = 6.1 Hz), 129.8, 130.5, 159.3, 184.6 (t, J = 26.1 Hz); ^{19}F NMR ($CDCl_3$): δ –111.96 (s, 2F), –107.66 (s, 2F); IR (neat): 2966, 1700, 1604, 1412, 1366, 1289, 1158, 996, 931, 887, 859, 788 cm^{-1}; HRMS(FAB) Calcd for $C_{15}H_{16}BrF_4O$ $[M+H]^+$: 367.0321, Found 367.0322.

4.5.4. 4-Bromo-1-(4-bromophenyl)−2,2,3,3-tetrafluoropent-4-en-1-one *(6e)*

Yield: 79% (0.15 g, 0.39 mmol); Pale yellow oil (Hexane/Benzene = 10/1, R_f = 0.33); ^{1}H NMR ($CDCl_3$): δ 6.21 (dm, J = 3.0 Hz, 1H), 6.50 (d, J = 3.0 Hz, 1H), 7.65–7.69 (m, 2H), 7.95 (d, J = 8.5 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 111.9 (tt, J = 267.2, 37.3 Hz), 112.7 (tt, J = 256.5, 31.7 Hz), 117.9 (t, J = 28.8 Hz), 127.6 (t, J = 6.3 Hz), 131.0, 131.2, 131.9, 132.4, 184.4 (t, J = 26.5 Hz); ^{19}F NMR ($CDCl_3$): δ –112.27 (s, 2F), –107.63 (s, 2F); IR (neat): 3121, 2918, 1707, 1584, 1401, 1285, 1159, 1075, 1013, 882, 758 cm^{-1}; HRMS (FAB) calcd for $C_{11}H_8BrF_4O$ $[M+H]^+$: 388.8800, found 388.8792.

4.5.5. 4-Bromo-2,2,3,3-tetrafluoro-1-(4-trifluoromethylphenyl)pent-4-en-1-one *(6f)*

Yield: 51% (0.09 g, 0.25 mmol), Pale yellow oil (Hexane/Benzene = 1/1, R_f = 0.78); ^{1}H NMR ($CDCl_3$): δ 6.20 (m, 1H), 6.51 (d, J = 3.0 Hz, 1H), 7.76 (d, J = 8.5 Hz, 2H), 8.19 (d, J = 8.3 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 111.9 (tt, J = 267.0, 37.5 Hz), 112.8 (tt, J = 256.3, 31.6 Hz), 117.7 (t, J = 28.7 Hz), 123.5 (q, J = 272.8 Hz), 126.0 (q, J = 3.4 Hz), 127.7 (t, J = 6.2 Hz), 130.8, 135.2, 136.1 (q, J = 33.0 Hz), 184.7 (t, J = 26.9 Hz); ^{19}F NMR ($CDCl_3$): δ –112.23 (s, 2F), –107.31 (s, 2F), –63.93 (m, 3F); IR (neat): 3123, 2928, 1714, 1626, 1511, 1413, 1329, 1220, 1137, 1069, 888, 751 cm^{-1}; MS (FAB): m/z 299 ($CH_2{=}C^+CF_2$ $CF_2CO(C_6H_4CF_3$-p), 11), 173 (p-$CF_3C_6H_4C^+{\equiv}O$, 90), 145 ($[p$-$CF_3C_6H_4]^+$, 29).

4.5.6. 4-Bromo-2,2,3,3-tetrafluoro-1-(2-methylphenyl)pent-4-en-1-one (6 h)

Yield: 60% (0.10 g, 0.31 mmol); Pale yellow oil (Hexane/Benzene = 10/1, R_f = 0.33); ^{1}H NMR ($CDCl_3$): δ 2.47 (s, 3H), 6.19 (dm, J = 2.9 Hz, 1H), 6.49 (d, J = 2.9 Hz, 1H), 7.29–7.33 (m, 2H), 7.46–7.50 (m, 1H), 7.80 (d, J = 8.0 Hz, 1H); ^{13}C NMR ($CDCl_3$): δ 21.1, 111.1 (tt, J = 268.7, 36.9 Hz), 113.0 (tt, J = 256.4, 32.0 Hz), 118.3 (t, J = 28.9 Hz), 125.7, 127.2 (t, J = 6.4 Hz), 129.5 (t, J = 5.6 Hz), 132.2, 132.7, 133.0, 140.3, 188.5 (t, J = 26.3 Hz); ^{19}F NMR ($CDCl_3$): δ –111.94 (s, 2F), –107.45 (s, 2F); IR (neat): 3067, 2971, 1709, 1624, 1601, 1571, 1457, 1295, 1154, 1076, 862, 736 cm^{-1}; HRMS (FAB) calcd for $C_{12}H_{10}F_4O$ $[M]^+$: 324.9851, found 324.9842.

4.5.7. 4-Bromo-2,2,3,3-tetrafluoro-1-(3-methylphenyl)pent-4-en-1-one *(6i)*

Yield: 64% (0.12 g, 0.36 mmol); Pale yellow oil (Hexane/EtOAc = 50/1, R_f = 0.31); ^{1}H NMR ($CDCl_3$): δ 2.43 (s, 3H), 6.20 (dm, J = 3.0 Hz, 1H), 6.50 (d, J = 3.0 Hz, 1H), 7.40 (t, J = 8.2 Hz, 1H), 7.49 (d, J = 7.6 Hz, 1H), 7.89–7.91 (m, 2H); ^{13}C NMR ($CDCl_3$): δ 21.4, 111.9 (tt, J = 267.3, 36.8 Hz), 112.7 (tt, J = 256.6, 31.6 Hz), 118.2 (t, J = 28.8 Hz), 127.3 (t, J = 6.1 Hz), 127.7 (t, J = 4.1 Hz), 128.8, 130.8, 132.4, 135.9, 138.9, 185.4 (t, J = 26.0 Hz); ^{19}F NMR ($CDCl_3$): δ –111.82 (s, 2F), –107.62 (s, 2F); IR (neat): 3386, 3121, 3033, 2927, 1704, 1626, 1604, 1585, 1423, 1288, 1152, 1082, 933, 829, 735 cm^{-1}; HRMS (FAB) calcd for $C_{12}H_{10}BrF_4O$ $[M+H]^+$: 324.9851, found 324.9839.

4.5.8. 4-Bromo-1-(2-bromophenyl)−2,2,3,3-tetrafluoropent-4-en-1-one *(6j)*

Yield: 54% (0.11 g, 0.27 mmol); Pale yellow oil (Hexane/EtOAc = 50/1, R_f = 0.36); ^{1}H NMR ($CDCl_3$): δ 6.20–6.21 (m, 1H), 6.50 (d, J = 3.0 Hz, 1H), 7.38–7.45 (m, 2H), 7.55–7.57 (m, 1H), 7.68–7.71 (m, 1H); ^{13}C NMR ($CDCl_3$): δ 110.2 (tt, J = 268.6, 37.4 Hz), 112.9 (tt, J = 256.5, 32.0 Hz), 117.8 (t, J = 28.7 Hz), 120.5, 127.3, 127.6 (t, J = 6.3 Hz), 129.1, 133.2, 134.2, 135.7, 188.4 (t, J = 28.4 Hz); ^{19}F NMR ($CDCl_3$): δ –113.66 (s, 2F), –107.53 (s, 2F); IR (neat): 3120, 2857, 1734, 1624, 1587, 1469, 1433, 1288, 1227, 1153, 1082, 933, 878, 743 cm^{-1}; MS (FAB): m/z 205 ($CH_2{=}CBrCF_2C^+F_2$, 30), 183 (o-$BrC_6H_4C^+{\equiv}O$, 37), 155 ($[o$-$BrC_6H_4]^+$, 10).

4.5.9. 4-Bromo-1-(3-bromophenyl)−2,2,3,3-tetrafluoropent-4-en-1-one *(6k)*

Yield: 55% (0.12 g, 0.31 mmol); Pale yellow oil (Hexane only, R_f = 0.28); ^{1}H NMR ($CDCl_3$): δ 6.21–6.22 (m, 1H), 6.51 (d, J = 3.0 Hz, 1H), 7.41 (t, J = 8.0 Hz, 1H), 7.80 (dm, J = 8.0 Hz, 1H), 8.02 (d, J = 7.9 Hz, 1H), 8.20 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 111.7 (tt, J = 267.4, 37.4 Hz), 112.6 (tt, J = 256.7, 31.6 Hz), 117.8 (t, J = 28.7 Hz), 123.2, 127.6 (t, J = 6.3 Hz), 129.0, 130.5, 133.2, 134.0, 138.0, 184.2 (t, J = 26.8 Hz); ^{19}F NMR ($CDCl_3$): δ –112.22 (s, 2F), –107.60 (s, 2F); IR (neat): 3397, 3119, 3069, 2929, 1710, 1625, 1588, 1565, 1473, 1421, 1275, 1161, 999, 879, 746 cm^{-1}; HRMS (FAB) calcd for $C_{11}H_7Br_2F_4O$ $[M+H]^+$: 388.8800, found 388.8811.

4.5.10. 4-Bromo-2,2,3,3-tetrafluoro-1-(furan-2-yl)pent-4-en-1-one (**6l**)

Yield: 61% (0.08 g, 0.26 mmol); Pale yellow oil (Hexane/Benzene = 1/1, $R_f = 0.42$); ^1H NMR ($CDCl_3$): δ 6.19 (dm, $J = 3.0$ Hz, 1H), 6.49 (d, $J = 3.0$ Hz 1H), 6.66 (dd, $J = 3.8, 1.6$ Hz, 1H), 7.51–7.52 (m, 1H), 7.80 (m, 1H); ^{13}C NMR ($CDCl_3$): δ 111.2 (tt, $J = 267.0, 37.4$ Hz), 112.5 (tt, $J = 256.7, 32.1$ Hz), 113.3 (Ar), 117.7 (t, $J = 28.8$ Hz), 124.7–124.8, 127.6 (t, $J = 6.1$ Hz), 148.6, 150.0, 172.5 (t, $J = 27.8$ Hz); ^{19}F NMR ($CDCl_3$): δ –116.10 (s, 2F), –108.27 (s, 2F); IR (neat): 3148, 2926, 1688, 1558, 1459, 1396, 1158, 1086, 930, 845, 772 cm^{-1}; HRMS (FAB) calcd for $C_9H_7F_4O_2$ $[M + H]^+$: 300.9487, found 300.9484.

4.5.11. 4-Bromo-2,2,3,3-tetrafluoro-1-(thiophen-2-yl)pent-4-en-1-one (6 m)

Yield: 81% (0.12 g, 0.36 mmol); Yellow oil (Hexane/Benzene = 1/1, $R_f = 0.58$); ^1H NMR ($CDCl_3$): δ 6.19–6.20 (m, 1H), 6.50 (d, $J = 3.0$ Hz, 1H), 7.23 (t, $J = 4.5$ Hz, 1H), 7.87 (d, $J = 5.0$ Hz, 1H), 8.00–8.01 (m, 1H); ^{13}C NMR ($CDCl_3$): δ 111.6 (tt, $J = 266.7, 37.1$ Hz), 112.6 (tt, $J = 256.7, 31.9$ Hz), 117.9 (t, $J = 28.8$ Hz), 127.6 (t, $J = 5.8$ Hz), 129.2, 136.9, 137.8, 138.9, 177.9 (t, $J = 27.4$ Hz); ^{19}F NMR ($CDCl_3$): δ –114.14 (s, 2F), –108.03 (s, 2F); IR (neat): 3116, 1677, 1625, 1512, 1411, 1358, 1219, 1166, 1068, 980, 823, 731 cm^{-1}; HRMS (FAB) calcd for $C_9H_7F_4OS$ $[M + H]^+$: 316.9259, found 316.9261.

4.5.12. (1E)−4-Bromo-4,4,5,5-tetrafluoro-1-phenylhept-1,6-diene-3-one (**6n**)

Yield: 54% (0.10 g, 0.30 mmol); Pale yellow oil (Hexane/CH_2Cl_2 = 4/1, $R_f = 0.28$); ^1H NMR ($CDCl_3$): δ 6.15–6.16 (m, 1H), 6.49 (d, $J = 3.1$ Hz, 1H), 7.16 (d, $J = 15.8$ Hz, 1H), 7.38–7.47 (m, 3H), 7.59 (d, $J = 6.9$ Hz, 2H), 7.90 (d, $J = 15.8$ Hz); ^{13}C NMR ($CDCl_3$): δ = 110.8 (tt, $J = 265.5, 37.1$ Hz), 112.7 (tt, $J = 255.6, 32.3$ Hz), 117.7 (t, $J = 28.9$ Hz), 117.9, 127.4 (t, $J = 6.0$ Hz), 129.1, 129.2, 132.1, 133.5, 148.7, 183.7 (t, $J = 26.1$ Hz); ^{19}F NMR ($CDCl_3$): δ –120.09 (s, 2F), –108.42 (s, 2F); IR (neat): 3063, 2928, 1801, 1708, 1607, 1576, 1451, 1340, 1151, 1087, 944, 745 cm^{-1}; HRMS (FAB) calcd for $C_{13}H_{11}F_4O$ $[M + H]^+$: 336.9851, found 336.9857.

4.5.13. 4-Bromo-2,2,3,3-tetrafluorododec-4-en-1-one (**6o**)

Yield: 17% (0.03 g, 0.09 mmol); Pale yellow oil (Hexane only, $R_f = 0.33$); ^1H NMR ($CDCl_3$): δ 0.88 (t, $J = 6.9$ Hz), 1.25–1.31 (m, 8H), 1.65 (q, $J = 7.2$ Hz, 2H), 2.75 (t, $J = 7.2$ Hz, 2H), 6.18 (dm, $J = 3.0$ Hz, 1H), 6.46 (d, $J = 2.9$ Hz, 1H); ^{13}C NMR ($CDCl_3$): δ 14.2, 22.7, 28.9, 29.1, 31.7, 38.5, 107.25–115.66 (m, CF_2CF_2), 117.81 (t, $J = 29.2$ Hz), 127.37 (t, $J = 6.4$ Hz), 196.3 (t, $J = 25.5$ Hz) (one sp^3-hybridized carbon was overlapped with another carbon); ^{19}F NMR ($CDCl_3$): δ –119.75 (s, 2F), –108.66 (s, 2F); IR (neat): 2929, 2857, 1752, 1625, 1351, 1238, 1154, 1084, 931 cm^{-1}; MS (FAB): m/z 127 (n-$C_7H_{15}C^+\equiv O$, 15).

4.6. Typical procedure for improved Cu(I)-catalyzed cross coupling reaction of organozinc reagent 3-Zn with iodobenzene

In a two-necked round-bottomed flask, equipped with a teflon-coated stirrer bar and reflux condenser, was added iodobenzene (0.051 g 0.25 mmol), Cu_2O (0.012 g, 0.08 mmol), pyridine (0.006 ml, 0.075 mmol), DMPU (2 ml), and a CH_3CN solution of **3-Zn** (0.38 M, 2.00 mL, 0.76 mmol), and the mixture was stirred at 100 °C for 24 h. After being cooled to room temperature, the resulting solution was subjected to a short pad of silica gel using hexane as an eluent. The filtrate was concentrated in *vacuo* to give the crude materials, which was purified by silica gel column chromatography to provide the corresponding compound **8a** in 28% yield.

4.6.1. 2-Bromo-3,3,4,4-tetrafluoro-4-phenylbut-1-ene (**8a**)

Yield: 28% (0.02 g, 0.07 mmol); Pale yellow oil (Hexane only, $R_f = 0.56$); ^1H NMR ($CDCl_3$): δ 6.11 (m, 1H), 6.34 (d, $J = 2.6$ Hz, 1H), 7.45–7.49 (m, 2H), 7.51–7.55 (m, 1H), 7.58–7.60 (m, 2H); ^{13}C NMR ($CDCl_3$): δ 113.3 (tt, $J = 255.4, 36.8$ Hz), 116.4 (tt, $J = 254.6, 35.2$ Hz), 119.2 (t, $J = 29.8$ Hz) 126.8–127.0 (m), 127.0 (tm, $J = 6.6$ Hz), 128.4, 130.5 (t, $J = 24.6$ Hz), 131.4; ^{19}F NMR ($CDCl_3$): δ –110.56 (s, 2F), –109.42 (s, 2F); IR (neat): 3070, 2927, 2856, 1625, 1607, 1499, 1455, 1381, 1027, 859 cm^{-1}; MS (FAB): m/z 205 ($CH_2{=}CBrCF_2C^+F_2$, 14), 155 ($CH_2{=}CBrC^+F_2$, 27), 127 ($C_6H_5C^+F_2$, 98).

4.6.2. 2-Bromo-3,3,4,4-tetrafluoro-4-(4-methylphenyl)but-1-ene (**8b**)

Yield: 50% (0.04 g, 0.13 mmol); Pale yellow oil (Hexane only, $R_f = 0.50$); ^1H NMR ($CDCl_3$): δ 2.42 (s, 3H), 6.11 (m, 1H), 6.34 (d, $J = 2.3$ Hz, 1H), 7.28 (d, $J = 7.5$ Hz, 2H), 7.48 (d, $J = 8.0$ Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 21.5 (s, 3H), 113.3 (tt, $J = 255.2, 36.8$ Hz), 116.6 (tt, $J = 254.6, 34.9$ Hz), 119.3 (t, $J = 30.0$ Hz), 126.8 (t, $J = 6.3$ Hz), 127.0 (t, $J = 6.2$ Hz), 127.6 (t, $J = 24.9$ Hz), 129.1, 141.7; ^{19}F NMR ($CDCl_3$): δ –110.12 (s, 2F), –109.40 (s, 2F); IR (neat): 3042, 2927, 2858, 1625, 1518, 1410, 1291, 1225, 1141, 1080, 936, 818, 755 cm^{-1}; MS (FAB): m/z 205 ($CH_2{=}CBrCF_2C^+F_2$, 11), 155 ($CH_2{=}CBrC^+F_2$, 25), 141 (p-$CH_3C_6H_4C^+F_2$, 23).

4.6.3. 2-Bromo-3,3,4,4-tetrafluoro-4-(4-methoxyphenyl)but-1-ene (**8c**)

Yield: 84% (0.07 g, 0.23 mmol); Yellow oil (Hexane only, $R_f = 0.24$); ^1H NMR ($CDCl_3$): δ 3.84 (s, 3H), 6.10 (m, 1H), 6.33 (d, $J = 2.6$ Hz, 1H), 6.96 (d, $J = 8.6$ Hz, 2H), 7.50 (d, $J = 8.8$ Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 55.5, 113.3 (tt, $J = 255.4, 37.8$ Hz), 113.8, 116.6 (tt, $J = 254.7, 35.2$ Hz), 119.4 (t, $J = 28.7$ Hz), 122.5 (t, $J = 24.8$ Hz), 126.7 (t, $J = 5.9$ Hz), 128.6 (t, $J = 6.2$ Hz), 161.9; ^{19}F NMR ($CDCl_3$): δ –109.48 (s, 2F), –109.46 (s, 2F); IR (neat): 2963, 2843, 1616, 1588, 1310, 1261, 1181, 1101, 1087, 932, 834, 756 cm^{-1}; HRMS (FAB) calcd for $C_{11}H_9BrF_4O$ $[M]^+$: 311.9773, found 311.9769.

4.6.4. 2-Bromo-4-(4-chlorophenyl)−3,3,4,4-tetrafluorobut-1-ene (**8d**)

Yield: 57% (0.05 g, 0.15 mmol); Pale yellow oil (Hexane only, $R_f = 0.56$); ^1H NMR ($CDCl_3$): δ 6.13 (m, 1H), 6.36 (d, $J = 2.4$ Hz, 1H), 7.45 (d, $J = 8.4$ Hz, 2H), 7.52 (d, $J = 8.4$ Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 113.1 (tt, $J = 255.5, 39.0$ Hz), 116.2 (tt, $J = 254.9, 35.5$ Hz), 118.9 (t, $J = 29.8$ Hz), 127.1 (t, $J = 6.0$ Hz), 128.5 (t, $J = 6.4$ Hz), 128.9, 129.0 (t, $J = 25.1$ Hz), 137.9; ^{19}F NMR ($CDCl_3$): δ –110.49 (s, 2F), –109.36 (s, 2F); IR (neat): 2928, 2856, 1625, 1605, 1496, 1406, 1292, 1233, 1141, 1095, 936, 827, 766 cm^{-1}; MS (FAB): m/z 161 (p-$ClC_6H_4C^+F_2$, 10), 155 ($CH_2{=}CBrC^+F_2$, 37), 111 ($[p$-$ClC_6H_4]^+$, 27).

4.6.5. Ethyl 4-(3-bromo-1,1,2,2-tetrafluorobut-3-en-1-yl)benzoate (**8e**)

Yield: 71% (0.07 g, 0.19 mmol); White solid; M.p.: 39–40 °C (Hexane/EtOAc = 30/1, $R_f = 0.19$); ^1H NMR ($CDCl_3$): δ 1.40 (t, $J = 7.1$ Hz, 3H), 4.40 (q, $J = 7.2$ Hz, 2H), 6.11 (m, 1H), 6.34 (d, $J = 2.7$ Hz, 1H), 7.65 (d, $J = 8.3$ Hz, 2H), 8.13 (d, $J = 8.3$ Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 14.4, 61.6, 113.1 (tt, $J = 255.8, 36.0$ Hz), 116.1 (tt, $J = 255.3, 35.5$ Hz), 118.8 (t, $J = 29.7$ Hz), 127.2 (t, $J = 6.0$ Hz), 129.6, 134.5 (t, $J = 24.6$ Hz), 165.7, (one sp^2-hybridized carbon was overlapped with another carbon); ^{19}F NMR ($CDCl_3$): δ –110.96 (s, 2F), –109.32 (s, 2F); IR (neat): 2988, 2905, 1952, 1716, 1625, 1412, 1296, 1220, 1138, 1025, 924, 853, 716 cm^{-1}; HRMS (FAB) calcd for $C_{13}H_{12}BrF_4O_2$ $[M + H]^+$: 354.9957, found 354.9952.

4.6.6. 2-Bromo-3,3,4,4-tetrafluoro-4-(2-methylphenyl)but-1-ene (**8f**)

Yield: 61% (0.047 g, 0.16 mmol); Pale yellow oil (Hexane only, $R_f = 0.56$); ^1H NMR ($CDCl_3$): δ 2.51 (t, $J = 3.1$ Hz, 3H), 6.15 (m, 1H), 6.39 (d, $J = 2.6$ Hz, 1H), 7.24–7.29 (m, 2H), 7.39 (t, $J = 7.4$ Hz, 1H), 7.50 (d, $J = 7.8$ Hz, 1H); ^{13}C NMR ($CDCl_3$): δ 20.8, 113.9 (tt, $J = 258.1, 38.0$ Hz), 117.9 (tt, $J = 255.5, 36.6$ Hz), 119.5 (t, $J = 30.1$ Hz), 125.7, 126.8 (t, $J = 6.0$ Hz), 128.5 (t, $J = 22.6$ Hz), 128.8 (t, $J = 8.8$ Hz), 131.2, 132.4, 138.1; ^{19}F NMR ($CDCl_3$): δ –108.28 (s, 2F), –105.61 (s, 2F); IR (neat): 3067, 2936, 1731, 1625, 1606, 1461, 1298, 1218, 1143, 1083, 934, 752 cm^{-1}; MS (FAB): m/z 155 ($CH_2{=}CBrC^+F_2$, 38), 91 ($[o$-$CH_3C_6H_5]^+$, 86).

4.6.7. 2-Bromo-3,3,4,4-tetrafluoro-4-(3-methylphenyl)but-1-ene (8 g)

Yield: 44% (0.035 g, 0.12 mmol); Pale yellow oil (Hexane only, R_f =

0.58); ^{1}H NMR ($CDCl_3$): δ 2.41 (s, 3H), 6.11 (m, 1H), 6.35 (d, J = 2.6 Hz, 1H), 7.32–7.35 (m, 2H), 7.37–7.39 (m, 2H); ^{13}C NMR ($CDCl_3$): δ 21.5, 113.3 (tt, J = 255.8, 37.2 Hz), 116.5 (tt, J = 254.7, 34.8 Hz), 119.5 (t, J = 30.1 Hz), 124.2 (t, J = 6.3 Hz), 126.7 (t, J = 6.0 Hz), 127.6 (t, J = 5.9 Hz), 128.3, 130.5 (t, J = 24.4 Hz), 132.1, 138.3; ^{19}F NMR ($CDCl_3$): δ –110.27 (s, 2F), –109.24 (s, 2F); IR (neat): 2927, 2856, 1729, 1626, 1490, 1230, 1191, 1133, 1086, 1049, 962, 835, 754 cm^{-1}; MS (FAB): *m/z* 155 ($CH_2{=}CBrC^+F_2$, 21), 141 (p-$CH_3C_6H_4C^+F_2$, 46), 91 ([m-$CH_3C_6H_5$]$^+$, 57).

4.6.8. 2-Bromo-4-(2-chlorophenyl)−3,3,4,4-tetrafluorobut-1-ene (8 h)

Yield: 67% (0.10 g, 0.30 mmol); Pale yellow oil (Hexane only, R_f = 0.50); ^{1}H NMR ($CDCl_3$): δ 6.17 (dm, J = 2.7 Hz, 1H), 6.43 (d, J = 2.7 Hz, 1H), 7.36 (tm, J = 7.3 Hz, 1H), 7.42–7.49 (m, 2H), 7.61 (dm, J = 8.0 Hz, 1H); ^{13}C NMR ($CDCl_3$): δ 113.6 (tt, J = 256.1, 36.5 Hz), 116.3 (tt, J = 257.2, 37.1 Hz), 119.1 (t, J = 29.7 Hz, CH_2CBr), 126.7, 127.1 (t, J = 6.3 Hz), 128.0 (t, J = 23.5 Hz), 130.4 (t, J = 8.2 Hz), 132.1, 132.5, 133.6 (t, J = 2.7 Hz); ^{19}F NMR ($CDCl_3$): δ –107.76 (s, 2F), –106.60 (s, 2F); IR (neat): 3121, 3076, 2930, 1723, 1625, 1596, 1477, 1436, 1294, 1220, 1155, 1098, 939, 763 cm^{-1}; MS (FAB): *m/z* 161 (o-$ClC_6H_4C^+F_2$, 56), 155 ($CH_2{=}CBrC^+F_2$, 30), 111 ([o-ClC_6H_5]$^+$, 27).

4.6.9. 2-Bromo-4-(3-chlorophenyl)−3,3,4,4-tetrafluorobut-1-ene (8i)

Yield: 54% (0.051 g, 0.16 mmol); Pale yellow oil (Hexane only, R_f = 0.56); ^{1}H NMR ($CDCl_3$): δ 6.14 (m, 1H), 6.37 (m, 1H), 7.39–7.58 (m, 4H); ^{13}C NMR ($CDCl_3$): δ 113.1 (tt, J = 255.6, 36.6 Hz), 115.7 (tt, J = 255.4, 35.7 Hz), 118.8 (t, J = 29.6 Hz), 125.3 (t, J = 6.1 Hz), 127.2 (t, J = 6.4 Hz), 127.4 (t, J = 6.2 Hz), 129.8, 131.7, 132.3 (t, J = 24.6 Hz), 134.7; ^{19}F NMR ($CDCl_3$): δ –110.53 (s, 2F), –109.19 (s, 2F); IR (neat): 3076, 2929, 2856, 1625, 1580, 1480, 1428, 1291, 1224, 1144, 1089, 1046, 940, 886, 794 cm^{-1}; MS (FAB): *m/z* 155 ($CH_2{=}CBrC^+F_2$, 54), 111 ([m-ClC_6H_5]$^+$, 29).

4.6.10. Ethyl 2-(3-bromo-1,1,2,2-tetrafluorobut-3-en-1-yl)benzoate (8j)

Yield: 74% (0.079 g, 0.22 mmol); Yellow oil (Hexane/EtOAc = 30/1, R_f = 0.14); ^{1}H NMR ($CDCl_3$): δ 1.36 (t, J = 7.2 Hz, 3H), 4.36 (q, J = 7.2 Hz, 2H), 6.16 (dm, J = 2.8 Hz, 1H), 6.43 (d, J = 2.8 Hz, 1H), 7.49–7.64 (m, 4H); ^{13}C NMR ($CDCl_3$): δ 14.1, 62.0, 113.2 (tt, J = 256.1, 37.3 Hz), 116.6 (tt, J = 256.0, 37.2 Hz), 119.0 (t, J = 29.6 Hz), 127.1 (t, J = 6.2 Hz), 127.4, 128.6, 128.8 (tt, J = 7.3, 2.2 Hz), 129.8, 131.3, 133.8 (t, J = 3.4 Hz, Ar), 168.37; ^{19}F NMR ($CDCl_3$): δ –107.13 (s, 2F), –104.26 (s, 2F); IR (neat): 2985, 1735, 1625, 1446, 1368, 1300, 1218, 1149, 1056, 940, 919, 764, 681 cm^{-1}; HRMS (FAB) calcd for $C_{13}H_{12}BrF_4O_2$ [$M + H$]$^+$: 354.9957, found 354.9962.

4.6.11. 2-(3-Bromo-1,1,2,2-tetrafluorobut-3-en-1-yl)thiophene (8k)

Yield: 59% (0.053 g, 0.18 mmol); yellow oil (Hexane only, R_f = 0.33); ^{1}H NMR ($CDCl_3$): δ 6.14 (dm, J = 2.8 Hz, 1H), 6.40 (d, J = 2.7 Hz, 1H), 7.09–7.12 (m, 1H), 7.42 (dm, J = 3.1 Hz, 1H), 7.53 (dd, J = 5.0 Hz, 1.2 Hz, 1H); ^{13}C NMR ($CDCl_3$): δ 113.0 (tt, J = 255.7, 36.8 Hz), 115.5 (tt, J = 253.7, 36.1 Hz), 118.9 (t, J = 29.9 Hz), 127.1, 127.2, 129.4, 129.9 (t, J = 5.2 Hz), 131.5 (t, J = 29.3 Hz); ^{19}F NMR ($CDCl_3$): δ –109.10 (s, 2F), –100.39 (s, 2F); IR (neat): 3118, 2928, 2855, 1730, 1625, 1534, 1432, 1360, 1276, 1229, 1118, 1087, 1016, 931, 856, 715 cm^{-1}; MS (FAB): *m/z* 155 ($CH_2{=}CBrC^+F_2$, 82), 83 ([C_4H_3S]$^+$, 14).

4.6.12. 3-(3-Bromo-1,1,2,2-tetrafluorobut-3-en-1-yl)pyridine (8l)

Yield: 53% (0.047 g, 0.17 mmol); Yellow oil (Hexane/EtOAc = 5/1, R_f = 0.33); ^{1}H NMR ($CDCl_3$): δ 6.15 (dm, J = 2.8 Hz, 1H), 6.39 (d, J = 2.8 Hz, 1H), 7.41 (dd, J = 7.5 Hz, 4.9 Hz, 1H), 7.88 (d, J = 8.0 Hz), 8.77 (d, J = 4.6 Hz); ^{13}C NMR ($CDCl_3$): δ 113.0 (tt, J = 255.4, 36.2 Hz), 115.70 (tt, J = 255.5, 36.3 Hz), 118.4 (t, J = 29.6 Hz), 126.7 (t, J = 24.8 Hz), 127.4 (t, J = 5.9 Hz), 134.8 (t, J = 5.9 Hz), 148.2 (t, J = 6.5 Hz), 152.5; ^{19}F NMR ($CDCl_3$): δ –111.30 (s, 2F), –109.30 (s, 2F); IR (neat): 3047, 2928, 1724, 1625, 1596, 1483, 1426, 1295, 1226, 1143, 1091, 1022, 942, 809, 717 cm^{-1}; HRMS (FAB) calcd for $C_9H_7BrF_4N$ [$M + H$]$^+$: 283.9698, found 283.9699.

4.7. General procedure for the synthesis of 9a via Suzuki-Miyaura cross coupling reaction

In a two-necked round-bottomed flask, equipped with a teflon-coated stirrer bar, was charged with **6a** (0.093 g, 0.3 mmol), $PhB(OH)_2$ (0.073 g, 0.6 mmol), $Pd(PPh_3)_4$ (0.017 g, 5 mol%), Na_2CO_3 (0.064 g, 0.6 mmol), toluene (0.48 mL) and H_2O (0.12 mL). The mixture was stirred at 80 °C under argon for overnight. The mixture was diluted with diethyl ether and washed with water, then dried over anhydrous sodium sulfate, and then evaporated *in vacuo*. The **9a** was purified by silica gel column chromatography (0.059 g, 0.19 mmol, 64%).

4.7.1. 2,2,3,3-Tetrafluoro-1,4-diphenylpent-4-en-1-one (9a)

Yield: 64% (0.059 g, 0.19 mmol); Yellow oil (Hexane/CH_2Cl_2 = 30/1, R_f = 0.17); ^{1}H NMR ($CDCl_3$): δ 5.75 (s, 1H), 5.96 (s, 1H), 7.34–7.43 (m, 5H), 7.48 (t, J = 7.8 Hz, 2H), 7.64 (t, J = 7.4 Hz, 2H), 8.03 (d, J = 7.9 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 112.7 (tt, J = 266.5, 38.1 Hz), 115.7 (tt, J = 255.6, 32.4 Hz), 124.5 (t, J = 8.0 Hz), 128.3, 128.7, 128.8, 129.0, 130.4, 132.8, 134.8, 135.7, 139.8 (t, J = 21.8 Hz), 186.0 (t, J = 25.9 Hz); ^{19}F NMR ($CDCl_3$): δ –112.23 (m, 2F), –108.67 (s, 2F); IR (neat): 3061, 2927, 2856, 2368, 1891, 1704, 1598, 1449, 1281, 1154, 1031, 920, 849, 728 cm^{-1}; HRMS (FAB) calcd for $C_{17}H_{13}F_4O$ [$M + H$]$^+$: 309.0903, found 309.0907.

4.7.2. 2,2,3,3-Tetrafluoro-4-(4-methylphenyl)−1-phenylpent-4-en-1-one (9b)

Yield: 66% (0.064 g, 0.20 mmol); Yellow oil (Hexane only, R_f = 0.08); ^{1}H NMR ($CDCl_3$): δ 2.38 (s, 3H), 5.73 (m, 1H), 5.92 (brs, 1H), 7.17 (d, J = 7.9 Hz, 2H), 7.32 (d, J = 8.0 Hz, 2H), 7.48 (tm, J = 7.8 Hz, 2H), 7.65 (tm, J = 7.0 Hz, 1H) 8.04 (d, J = 7.3 Hz, 2H);^{13}C NMR ($CDCl_3$): δ 21.3, 112.7 (tt, J = 266.2, 38.3 Hz), 115.8 (tt, J = 254.7, 31.7 Hz), 124.0 (t, J = 8.6 Hz), 128.8, 128.8, 130.4, 132.8, 134.8, 138.6, 139.6 (t, J = 21.7 Hz), 186.0 (t, J = 26.2 Hz) (one sp^2-hybridized carbon was overlapped with another carbon); ^{19}F NMR ($CDCl_3$): δ –112.25 (s, 2F), –108.59 (s, 2F); IR (neat): 3032, 2925, 2368, 1909, 1703, 1598, 1450, 1327, 1281, 1154, 1077, 943, 852, 721 cm^{-1}; HRMS (FAB) calcd for $C_{18}H_{15}F_4O$ [$M + H$]$^+$: 323.1059, found 323.1068.

4.7.3. 4-(4-Chlorophenyl)−2,2,3,3-tetrafluoro-1-phenylpent-4-en-1-one (9c)

Yield: 56% (0.058 g, 0.17 mmol); Yellow oil (Hexane only, R_f = 0.16); ^{1}H NMR ($CDCl_3$): δ 5.73 (t, J = 1.4 Hz, 1H), 5.95 (brs, 1H), 7.33 (m, 4H), 7.48 (tm, J = 8.2 Hz, 2H), 7.65 (tm, J = 7.5 Hz, 1H), 8.02 (d, J = 7.8 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 112.6 (tt, J = 266.1, 37.9 Hz), 115.5 (tt, J = 255.1, 31.5 Hz), 125.0 (t, J = 8.6 Hz), 128.6, 128.8, 130.3, 130.4, 132.6, 134.1, 134.9, 134.9, 138.7 (t, J = 22.2 Hz), 185.9 (t, J = 26.3 Hz); ^{19}F NMR ($CDCl_3$): δ –112.23 (s, 2F), –109.05 (s, 2F); IR (neat): 3065, 2929, 2370, 1908, 1703, 1598, 1493, 1397, 1281, 1155, 1093, 950, 851, 739 cm^{-1}; HRMS (FAB) calcd for $C_{17}H_{12}ClF_4O$ [$M + H$]$^+$: 343.0513, found 343.0509.

4.8. General procedure for the synthesis of 10a via Suzuki-Miyaura cross coupling reaction

In a reaction vial, equipped with a teflon-coated stirrer bar, was charged with **8e** (0.108 g, 0.3 mmol), $PhB(OH)_2$ (0.074 g, 0.6 mmol), $Pd(PPh_3)_4$ (0.017 g, 5 mol%), Na_2CO_3 (0.064 g, 0.6 mmol), toluene (0.48 mL) and H_2O (0.12 mL). The mixture was stirred at 80 °C under argon for overnight. The mixture was diluted with diethyl ether and washed with water, then dried over anhydrous sodium sulfate, and then evaporated *in vacuo*. The **10a** was purified by silica gel column chromatography (0.076 g, 0.22 mmol, 72%).

4.8.1. Ethyl 4-(1,1,2,2-tetrafluoro-3-phenylbut-3-en-1-yl)benzoate (**10a**)

Yield: 72% (0.076 g, 0.22 mmol); Yellow solid; M.p.: 43–45 °C; (Hexane/EtOAc = 30/1, R_f = 0.28); ^{1}H NMR ($CDCl_3$): δ 1.41 (t, J = 7.2 Hz, 3H), 4.41 (q, 7.1 Hz, 2H), 5.69 (brs, 1H), 5.82 (s, 1H), 7.33–7.40 (m, 5H), 7.60 (d, J = 8.4 Hz, 2H), 8.10 (d, J = 8.6 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 14.4, 61.5, 116.1 (tt, J = 254.1, 36.2 Hz), 116.6 (tt, J = 254.4, 36.2 Hz), 124.3 (t, J = 8.5 Hz), 127.2 (t, J = 6.1 Hz), 128.2, 128.5, 128.7, 129.4, 133.0, 135.3 (t, J = 24.9 Hz), 136.3, 140.2 (t, J = 22.3 Hz), 165.8; ^{19}F NMR ($CDCl_3$): δ –110.59 (s, 2F), –109.73 (s, 2F); IR (neat): 3059, 2984, 2906, 2372, 1948, 1723, 1618, 1496, 1411, 1368, 1276, 1160, 1049, 943, 817, 720 cm^{-1}; MS (FAB): m/z 307 ($CH_2{=}C(C_6H_5)CF_2CF_2C_6H_4(C^+{\equiv}O$-$p)$, 95).

4.8.2. Ethyl 4-(1,1,2,2-tetrafluoro-3-(4-methylphenyl)but-3-en-1-yl) benzoate (**10b**)

Yield: 77% (0.085 g, 0.23 mmol); Yellow oil (Hexane/EtOAc = 30/1, R_f = 0.28); ^{1}H NMR ($CDCl_3$): δ 1.42 (t, J = 7.2 Hz, 3H), 2.38, 4.42 (q, J = 7.1 Hz, 2H), 5.66 (brs, 1H), 5.77 (s, 1H), 7.16 (d, J = 8.4 Hz, 2H), 7.28 (d, J = 8.0 Hz, 2H), 7.60 (d, J = 8.3 Hz, 2H), 8.10 (d, J = 8.2 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 14.4, 21.2, 61.4, 116.2 (tt, J = 254.0, 36.1 Hz), 116.7 (tt, J = 254.5, 36.1 Hz), 123.7 (t, J = 8.6 Hz), 127.2 (t, J = 6.2 Hz), 128.6, 129.0, 129.3, 133.0, 133.4, 135.4 (t, J = 24.9 Hz), 138.4, 140.1 (t, J = 22.2 Hz), 165.8; ^{19}F NMR ($CDCl_3$): δ –110.66 (s, 2F), –109.77 (s, 2F); IR (neat): 2984, 2368, 1908, 1612, 1542, 1411, 1368, 1225, 1184, 1050, 924, 858, 735 cm^{-1}; HRMS (FAB) calcd for $C_{20}H_{19}F_4O_2$ $[M+H]^+$: 367.1321, found 367.1309.

4.8.3. Ethyl 4-(3-(4-chlorophenyl)−1,1,2,2-tetrafluorobut-3-en-1-yl) benzoate (**10c**)

Yield: 74% (0.086 g, 0.22 mmol); Yellow solid, M.p.: 43−44 °C, (Hexane/EtOAc = 30/1, R_f = 0.25); ^{1}H NMR ($CDCl_3$): δ 1.40 (t, J = 7.2 Hz, 3H), 4.40 (q, J = 7.2 Hz, 2H), 5.67 (brs, 1H), 5.83 (s, 1H), 7.30 (m, 4H), 7.57 (d, J = 8.3 Hz, 2H), 8.09 (d, J = 8.4 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 14.4, 61.5, 115.9 (tt, J = 254.0, 36.3 Hz), 116.6 (tt, J = 254.4, 36.3 Hz), 124.7 (t, J = 8.3 Hz), 127.1 (t, J = 6.3 Hz), 128.5, 129.4, 130.1, 133.2, 134.7, 134.8, 135.1 (t, J = 24.9 Hz), 139.3 (t, J = 22.7 Hz), 165.7; ^{19}F NMR ($CDCl_3$): δ –110.56 (s, 2F), –109.86 (s, 2F); IR (neat): 2997, 2908, 1913, 1728, 1584, 1450, 1369, 1240, 1158, 1063, 955, 854, 736 cm^{-1}; HRMS (FAB) calcd for $C_{19}H_{16}ClF_4O_2$ $[M+H]^+$: 387.0775, found 387.0768.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jfluchem.2021.109781.

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